

STRAIN, PRESSURE, AND COMPLIANCE MODELING EXPANSION

Mechanical Compliance Measurement Concept

The wearable longitudinal penile physiology monitoring device may evaluate physiologic rigidity through measurement of mechanical compliance, deformation resistance, and contact force distribution between the wearable structure and underlying tissue. Mechanical compliance may represent a functional proxy for rigidity, allowing estimation of physiologic state without direct internal pressure measurement.

Compliance modeling may interpret how the wearable structure responds to expansion forces, including elastic deformation, displacement magnitude, rate-dependent stiffness behavior, and resistance to further dimensional change during erection progression.

Strain Sensing Expansion

Strain sensing elements may be incorporated into structural members of the wearable device to detect elongation, compression, bending, torsion, or multidirectional deformation. Strain sensors may include metallic foil gauges, semiconductor strain gauges, conductive polymer elements, printed flexible strain traces, fiber-optic strain sensors, capacitive strain sensors, or stretchable conductive networks integrated into flexible substrates.

Strain measurements may be obtained from single locations or distributed networks enabling circumferential deformation mapping and detection of asymmetric mechanical response patterns.

Pressure and Contact Force Modeling

Pressure sensing elements may measure localized force interactions between the wearable device and tissue. Sensors may include force-sensitive resistors, piezoresistive pressure sensors, capacitive pressure arrays, microfluidic pressure channels, compliant membrane sensors, or other force-detection technologies capable of measuring contact intensity and distribution.

Pressure distribution data may contribute to rigidity estimation by identifying uniform versus localized loading patterns, mechanical stability, and changes in tissue response during physiologic state transitions.

Compliance Curve and Response Modeling

Mechanical response curves may be generated by analyzing relationships between deformation and applied force over time. Compliance curves may characterize stiffness progression, saturation behavior, hysteresis effects, and recovery dynamics. Computational models may compare measured responses against individualized baseline profiles to derive physiologic interpretations.

Modeling approaches may include linear regression, nonlinear elasticity modeling, empirical curve fitting, probabilistic inference, or machine learning–based interpretation frameworks.

Hybrid Mechanical Sensing Architectures

Strain and pressure sensing modalities may operate independently or cooperatively with magnetic, optical, thermal, or inertial sensing channels to produce composite rigidity estimates. Hybrid mechanical sensing architectures may improve robustness against sensor noise, anatomical variability, and device positioning differences.

The compliance modeling embodiments described herein are illustrative and non-limiting and encompass present and future mechanical sensing technologies capable of characterizing longitudinal physiologic rigidity through wearable deformation analysis.